

Category theory

$$\mathbb{C} \setminus \{0\} \xrightarrow{z \mapsto z^2} \mathbb{C} \setminus \{0\}$$

Does this have an inverse?

Structural study
of topological spaces



equivalence

$$\mathbb{Z} \xrightarrow{n \mapsto 2n} \mathbb{Z}$$

Does this have an inverse?

Rather elementary
structural study of groups

Is $\partial\mathbb{D}$ a retract of $\mathbb{D}$?
Is there $\pi: \mathbb{D} \rightarrow \partial\mathbb{D}$ such that

$$\begin{array}{ccccc} \partial\mathbb{D} & \xrightarrow{\text{incl}} & \mathbb{D} & \xrightarrow{\pi} & \partial\mathbb{D} \\ & & & \nearrow \text{id}_{\partial\mathbb{D}} & \\ & & & & \end{array}$$



Is there a group homomorphism
 π_* such that

$$\begin{array}{ccccc} \mathbb{Z} & \xrightarrow{(\text{incl})_*} & \{0\} & \xrightarrow{\pi_*} & \mathbb{Z} \\ & & & \nearrow \text{id}_{\mathbb{Z}} & \\ & & & & \end{array}$$